Derivative Securities FNCE 30007

Tutorial 5

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October 22, 2026

Review

Notation:

- S_0: the current stock price.
- uS_0 (dS_0): the terminal stock price if the stock increases (decrease) in value.
- f_u (f_d): the value of the option if stock increases (decreases)

For a one-period binomial tree, we can create riskless portfolio with some Δ shares of stocks and short call. Payoff will be $\Delta uS_0 - f_u$ or $\Delta dS_0 - f_d$ depending on the price path. We can solve for $\Delta = \frac{f_u - f_d}{S_0(u-d)}$. Since we create a riskless portfolio, then $\Delta uS_0 - f_u = \Delta dS_0 - f_d \equiv \pi$.

The PV of the riskless payoff is $e^{-rT}\pi$. Then, we will solve for f_0 such that $\Delta S_0 - f_0 = e^{-rT}\pi \rightarrow f_0 = e^{-rT}(pf_u + (1-p)f_d)$, where $p = \frac{e^{rT} - d}{u - d}$

Review

We call p as the risk-neutral probability. No matter what is the physical probability of stock price increase in this one period binomial tree, the risk-neutral probability is always the same.
How to understand it (informally)?

1. If the conditional expected share price for this one-period binomial tree is the payoff of deposit S_0 into a risk-free bond to get $S_0 e^{rT}$ at maturity. $E[S_1|S_0] = p * uS_0 + (1-p)dS_0 = S_0 e^{rT}$.
2. Solve it, we get $p = \frac{e^{rT}-d}{u-d}$.
3. Therefore, as a risk-neutral investor who (does not require extra compensation for bearing risk) will use the risk-free rate to discount its future cash flow. Then, p is such probability of stock price increase in the view of a risk-neutral investor.

Tutorial Question

Q1: Consider the one-period binomial model for a stock price. Using the following results presented in lectures:

$$\Delta u S_0 - f_u = \pi$$

$$\Delta S_0 - f_0 = e^{-rT}\pi$$

Where f_u, f_d, and f_0 are the option values in the up state, down state, and time $t = 0$ respectively, and

$$\Delta = \frac{f_u - f_d}{S_0(u - d)}.$$

Show that:

$$f_0 = e^{-rT}\left(pf_u + (1 - p)f_d\right)$$

where

$$p = \frac{e^{rT} - d}{u - d}.$$

Tutorial Question

Q1: We have

$$\Delta S_0 - f_0 = e^{-rT}\pi \quad (1)$$

$$\Delta u S_0 - f_u = \pi \quad (2)$$

(2) $\Rightarrow$ (1):

$$e^{-rT}\left(\Delta u S_0 - f_u\right) = \Delta S_0 - f_0$$

where $\Delta = \frac{f_u - f_d}{S_0(u-d)}$

$$\Rightarrow e^{-rT}\left(\frac{f_u - f_d}{u-d}\cdot u - f_u\right) = \frac{f_u - f_d}{u-d} - f_0$$

$$\Rightarrow e^{-rT}\left(\frac{uf_u - uf_d - (u-d)f_u}{u-d}\right) = \frac{f_u - f_d}{u-d} - f_0$$

$$\Rightarrow e^{-rT}\left(\frac{f_u d - f_d u}{u-d}\right) = \frac{f_u - f_d}{u-d} - f_0$$

Tutorial Question

$$
\begin{aligned}
f_0 &= \frac{f_u - f_d}{u-d} - e^{-rT}\left(\frac{f_u d - f_d u}{u-d}\right) \\
&= \frac{1}{u-d}\left[(f_u - f_d) - e^{-rT}(f_u d - f_d u)\right] \\
&= \frac{1}{u-d}\left[f_u - f_d - e^{-rT} f_u d + e^{-rT} f_d u\right] \\
&= \frac{1}{u-d}\left[f_u(1 - e^{-rT} d) + f_d(e^{-rT} u - 1)\right] \\
&= \frac{1}{u-d}\left[f_u(e^{rT} - d) + f_d(u - e^{rT})\right] e^{-rT} \\
&= e^{-rT}\left(\frac{e^{rT} - d}{u-d} f_u + \frac{u - e^{rT}}{u-d} f_d\right)
\end{aligned}
$$

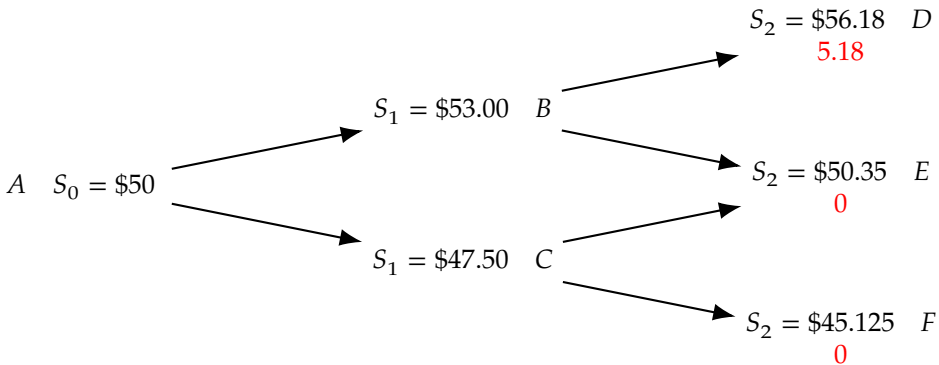
$$
\text{Let} \quad p = \frac{e^{rT} - d}{u-d} \Rightarrow f_0 = e^{-rT}\left(p f_u + (1-p) f_d\right)
$$

Tutorial Question

Q2: A stock price is currently $ 50. Over each of the next two three-month periods it is expected to go up by 6% or down by 5%. The continuously compounded risk-free interest rate is 5% p.a. with continuous compounding. What is the value of a six-month European call option with a strike price of 51?

Tutorial Question

Q2: A tree depicting the stock price path is below:



Tutorial Question

The risk-neutral probability of an up move in either three-month period is:

$$p = \frac{e^{0.05/4} - 0.95}{1.06 - 0.95} = 0.5689$$

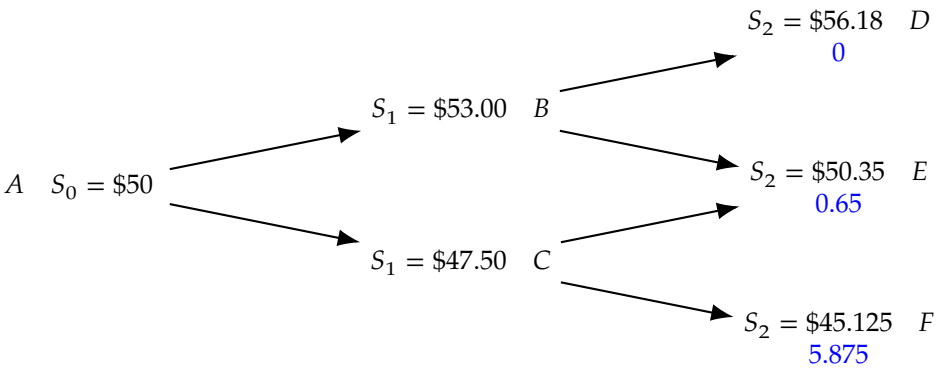
The option is worth \$5.18 at expiry Node D and 0 at Nodes E and F.
The option value is then:

$$f_0 = e^{-2\times 0.05/4} \times (0.5689^2 \times 5.18 + 0 + 0) = 1.6351 = \$1.64$$

Tutorial Question

Q3: For the situation considered in Question 2, what is the value of a six-month European put option with a strike price of \$51? Verify that the European call and European put prices satisfy put–call parity.

Tutorial Question



Tutorial Question

The value of the put option:

$$\frac{0.65 * 2 * 0.5689 * 0.4311 + 5.875 * 0.4311^2}{e^{0.05*6/12}} = 1.376$$

Recall Put-Call Parity:

$$c + Ke^{-rT} = p + S_0$$

we have $c = 1.635$, $K = 51$, $p = 1.376$, $S_0 = 50$, $T = 6/12$. Bring them together, we find the put call parity holds.

Tutorial Question

Q4: How do the answers to question 2 and 4 change if the probability of an up-move increases to 55% but all other details remain the same?

- The option price does not change if only the real-world probability of an up-move increases, as long as the current stock price, up and down factors (u and d), risk-free rate, and other inputs remain unchanged. This is because option pricing in the binomial model uses risk-neutral probabilities, not real-world probabilities. These depend only on the risk-free rate, u, and d; not on how likely an up-move is in reality.
- However, in reality, the current spot price of the stock is itself influenced by investors' beliefs about the real-world probabilities of future price movements. For example, if investors believe the stock is more likely to go up (higher physical probability of an up-move), they may be willing to pay more for it today, pushing the current spot price higher. Since option values depend on the current stock price, this could indirectly affect the option price. But within the model, with all inputs fixed, the physical probability has no effect.

Tutorial Question

Q5: A stock price is currently \$50. It is known that at the end of two months it will be either \$53 or \$48. The risk-free interest rate is 10% per annum with continuous compounding. What is the value of a two-month European call option with a strike price of \$49? Use no-arbitrage arguments.

Tutorial Question

At $T = \frac{2}{12}$ years the call will be worth \$4 if $S_T = 53$ and \$0 if $S_T = 48$. Consider a portfolio with $+\Delta$ shares and -1 call:

$$Value_T^{(u)} = 53\Delta - 4, \qquad Value_T^{(d)} = 48\Delta.$$

Make it riskless by equating the two terminal values:

$$48\Delta = 53\Delta - 4 \;\Rightarrow\; \Delta = 0.8, \quad V_T = 38.4.$$

Current cost:

$$V_0 = 0.8 \times 50 - f,$$

where f is the call value.
No-arbitrage $\Rightarrow$ the riskless portfolio must grow at the risk-free rate $r = 10\%$:

$$(0.8 \times 50 - f)\, e^{rT} = 38.4, \qquad T = \frac{2}{12}.$$

Solve for f:

$$\boxed{f = 40 - 38.4\, e^{-rT} = 2.23}.$$

Tutorial Question

Now, let's use the stardard formula to find the option value.
Let $u = 1.06,\ d = 0.96,\ r = 10\%,\ T = \frac{2}{12}$. Risk-neutral probability:

$$p = \frac{e^{rT} - d}{u - d} = \frac{e^{0.10\times 2/12} - 0.96}{1.06 - 0.96} = 0.5681.$$

Call price:

$$f = e^{-rT}[p \cdot 4 + (1-p)\cdot 0] = e^{-0.10\times 2/12} \times 0.5681 \times 4 = \boxed{2.23}.$$

Both approaches agree.